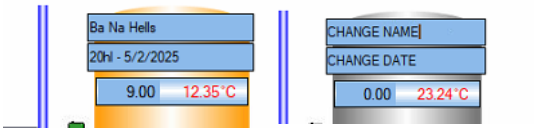
SOP Fermentation Ba Na Helles – Ba Na Brewhouse

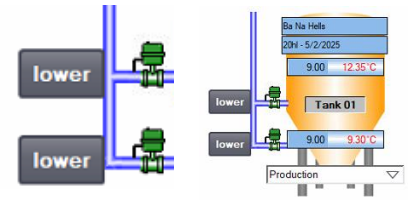
Step 1: Fill the **batch 1** into the FST (leave open the blue valve of the CIP line at the tank)

Step 2: Yeast pitching (Follow instructions from laboratory)

Step 3: Start fermentation program: "NEW Ba Na Hell" and change Name and Date on the graphics of the fermenter on BrauControl



Step 4: Adjust cooling valves and "cooling buttons" in BrauControl as shown in the pictures below





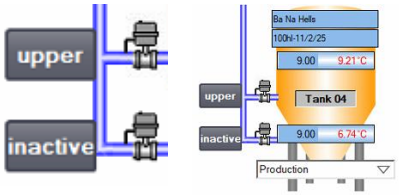
0% open

100% open

50% open

Step 5: Fill batch **2** and **3** into the FST

Step 6: After cooling of **batch 3** has finished, adjust cooling valves and "cooling buttons" like shown on the pictures below





100% open

50% open

0% open

Step 7: Fill batch **4, 5** and **6** into the **FST**

***Step 8*:** The day after filling of the FST has completed,

- Drain "**Cold Break"** from both, lower and upper outlets.
- Check if fermentation has started, by smelling at the CIP line of the tank. **ATTENTION! CO2 can be harmful, in big concentrations!**
- Close the blue valve of the CIP line at the tank and open the spunding valve completely so that CO2 can exit to the CO2 line

***Step 9*: Handover** from **Brewhouse department → Cellar department**

Step 10: **Follow fermentation! → Input data into fermentation protocol!**

- Take measurements of **gravity** and **ph** at the FST after every brewing day and after all batches have been filled into the tank
- Take measurements of **ph** until the result is **stable for 3 days**
- Take measurements of **gravity** until the result is **stable for 3 days**

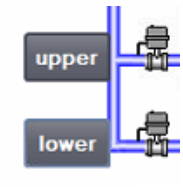
Step 11: close spunding valve when gravity has reached **4-5° Plato** and adjust to **1,2 bar**

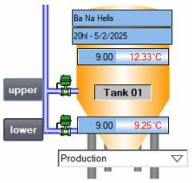
Step 12: After **gravity** has been stable for **3 days**, jump to the **next step** at **BrauControl: "Diacetyl Rest"** and measure **CO2**

- During this step, a beer tasting needs to be hold to find out if there is still Diacetyl left in the beer
  - If **"NO DIACETYL"** can be detected in the tasting→ jump to next step: **"Cooling to 0°C"**
  - If **"DIACETYL"** can still be detected in the tasting→leave the beer on fermenting temperature, repeat tasting on following days and jump to next step after no more diacetyl can be detected
  - The recipe in "BrauControl" automatically jumps into the next step (**"Cooling to 0°C")** after 3 days

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Step 13: Adjust cooling valves and "cooling buttons" in BrauControl as shown in the pictures below during the step **"Cooling to 0°C"**





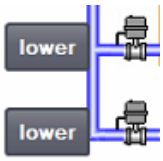


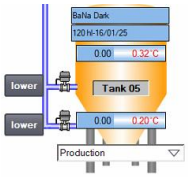
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Step 14: After the step **"Cooling to 0°C"** has completed, the recipe in BrauControl will jump automatically into the next step **"Storage 0°C"**→ When this step is reached, adjust cooling valves and "cooling buttons" in BrauControl as shown in the pictures below during the step







0% open

25% open

50% open

Step 15: During "Storage 0°C" step, follow the steps described below:

- If yeast needs to be harvested, harvest as soon as possible →After harvest, drain yeast every day, until no more yeast is coming out
- If no yeast needs to be harvested, drain yeast every day, until no more yeast is coming out
- Starting one week before date of centrifuge: Measure and adjust CO2 daily until the CO2 Level is inside the range

Step 16: One week before the beer is going to be centrifuged, the beer needs to be tasted to check, if the beer needs more time for storage.

- If the beer is "Ready for centrifuge", the beer can be centrifuged without another tasting.
- If the beer is still to young: "NOT ready for centrifuge", the beer needs to be tasted again at least one day before centrifuge. If it is still too young at the second tasting, the process of centrifuging needs to be delayed until the beer is "Ready for centrifuge"

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